EXPERIMENT - 9

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KRG 2A

Given two string str1 and str2, return shortest string that has both str1 and str2 as subsequences. If there are multiple valid strings, return any of them.

example:- str1 = "abac" str2 = "cab" output :- "cabac".

Thought Process → let's see about the length of shortest subsequence

$$n + m - lcs(str1, str2)$$

⇓

length of shortest subsequence.

lcs,

"a b a c" "c a b a c"

↑ ↑

let's build dp table first

```
if (str1[i] == str2[j])
    dp[i][j] = 1 + dp[i-1][j-1]
else
    dp[i][j] = max(dp[i-1][j], dp[i][j-1]);
```

After building DP Table,

```
i = n-1 , j = m-1          ans = "";
while (i>=0 && j>=0) {
    if (str1[i] == str2[j]) {
        ans+= str1[i]; i--; j--;
    } else if (dp[i-1][j] > dp[i][j-1]) {
        ans += str1[i]; i--;
    } else {
        ans += str2[j]; j--;
    }
}
while (i>=0) ans+= str1[i]; i--;
while (j>=0) ans += str2[j]; j--;
reverse(ans);
return ans;
```

to include strings extra as appear as subsequence is our requirement

as string is build in opposite fashion.